

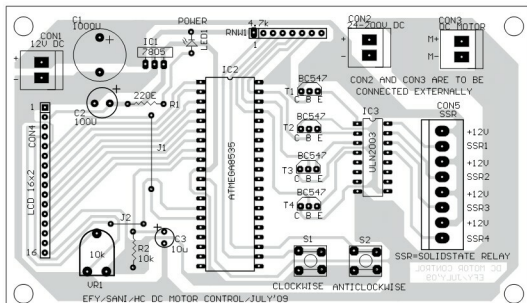


Fig. 7: Component layout for the PCB

tickclockwise direction.

If no message is shown on the LCD, vary VR1 and also check it for proper connections. If the message display is okay but the motor does not rotate, check the signals across the outputs of ULN2003. Also ensure proper connections to the relays.

EFY note. The source code of this project is included in this month's EFY-CD.

The author is a project manager at SDS Electronics, Haryana

DCMOTOR.C

```
//AVR Atmeg8535 based DC motor controller
#include <iom8535v.h>
#include <macros.h>
#include <math.h>

//////////prototypes//////////
void send_command(unsigned char command);
void lcd_init(void);
void send_data(unsigned char databyte);
void delay(unsigned char count1,unsigned char count2);
void switch_scan(void);

const unsigned char databyte[]={"DC MOTOR CONTROL SELECT DIRECTION CLOCKWISE ANTICLOCKWISE"};
//***** Function: Main *****/
void main(void)
{
    int i=0;
    DDWB=0xFF;
    DDRA=0xFF;
    DDRD=0xFF;
    PORTA=0x00;
    lcd_init();
    send_command(0x80);

    for(i=0;i<16;i++)
    {
        send_data(databyte[i]);
        delay(500,500);
        send_command(0x01);//clear lcd
        send_command(0x85);
        for(i=17;i<23;i++)
        {
            send_data(databyte[i]);
            send_command(0xc3);
            for(i=24;i<33;i++)
            {
                send_data(databyte[i]);
                while(1)
                {
                    switch_scan();
                }
            }
        }
    }
}
//***** Function: switch_scan *****/
void switch_scan(void)
{
    unsigned char i;
    if((PINC & 0x01)==0x01)
    {
        delay(10,50);
        while((PINC & 0x01)==0x01){
            send_command(0x01);
            send_command(0x82);
            for(i=34;i<43;i++){
                send_data(databyte[i]);
            }
            PORTA &= 0x0C;
            PORTA |= 0x03;
            if((PINC & 0x02)==0x02)
            {
                delay(10,50);
                while((PINC & 0x01)==0x01){
                    send_command(0x01);
                    send_command(0x82);
                    for(i=44;i<57;i++){
                        send_data(databyte[i]);
                    }
                    PORTA &= 0x03;
                    PORTA |= 0x0C;
                }
            }
        }
    }
}
//***** Function: LCD_init *****/
void lcd_init()
{
    delay(50,10);
    send_command(0x38);
    delay(50,10);
    send_command(0x0E);
    delay(50,10);
    send_command(0x01);
    delay(50,10);
    send_command(0x0c);
    delay(50,10);
    send_command(0x06);
    delay(50,10);
}
//***** Function: send command *****/
void send_command(unsigned char command)
{
    PORTD &= 0x01; // Clear RS
    delay(50,10);
    PORTD |= 0x02; // Set Enable
    delay(50,10);
    PORTB = command;
    delay(50,10); //clear enable
    PORTB &= 0x02;
    delay(50,10);
}
//***** Function: send data *****/
void send_data(unsigned char databyte)
{
    PORTD |= 0x01; //set RS
    delay(50,10);
    PORTD |= 0x02; // set Enable
    delay(50,10);
    PORTB = databyte;
    delay(50,10);
    PORTB &= 0x02; // clear Enable
    delay(50,10);
}
//***** Function: delay *****/
void delay(unsigned char count1,unsigned char count2)
{
    unsigned int i,j;
    for(i=0;i<count1;i++){
        for(j=0;j<count2;j++){
            ;
        }
    }
}
```